

A hypergeometric experiment

Randomly sample a fixed number without replacement from a finite population containing two categories.

How does this differ from independent binomial trials?

The group of interest

A synthetic box contains 7 red and 5 blue tokens. Select 4 uniformly without replacement; X counts red.

Which group is the group of interest?

The parameters

The textbook writes $X \sim H(r, b, n)$: r items of interest, b other items, n selected.

$$X \sim H(r, b, n)$$

For 7 red, 5 blue, and 4 draws, what are r , b , and n ?

Define X

A synthetic batch has 20 working and 5 faulty cables. Select 6 uniformly without replacement.

What random variable would count faulty cables in the sample?

A maximum count

A synthetic batch has 20 working and 5 faulty cables. Select 6 uniformly without replacement; X counts faulty.

Why can X not equal 6?

A minimum count

A synthetic box has 7 red and 3 blue tokens. Select 6 uniformly without replacement; X counts red.

What is the smallest possible value of X ?

The support

A synthetic box has 7 red and 3 blue tokens. Select 6 uniformly without replacement; X counts red.

What values can X take?

Conditional probability changes

A synthetic box has 7 red and 5 blue tokens.
Draw uniformly without replacement.

After a red draw, what is the probability the next token is red?

Count possible samples

A synthetic box has 7 red and 5 blue tokens.
Select 4 uniformly without replacement.

How many unordered four-token samples are possible?

Count favorable samples

A synthetic box has 7 red and 5 blue tokens.
Select 4 uniformly without replacement.

How many samples contain exactly 2 red and 2 blue tokens?

An exact probability

Synthetic box: 7 red and 5 blue tokens. Draw 4 uniformly without replacement; X counts red.

$$P(X = x) = \frac{\binom{r}{x} \binom{b}{n-x}}{\binom{r+b}{n}}$$

What hypergeometric probability gives $P(X = 2)$?

At most one

A synthetic batch has 20 working and 5 faulty cables. Select 6 uniformly without replacement; X counts faulty.

Which probabilities add to $P(X \leq 1)$?

A hypergeometric upper tail

Synthetic batch: 7 red, 5 blue tokens. Select 4 uniformly without replacement; X counts red.

What is $P(X > 2)$, using the hypergeometric formula or software?

An expected count

The expected hypergeometric count is sample size times the population proportion of interest.

$$\mu = n \frac{r}{r + b}$$

For 4 draws from 7 red and 5 blue tokens, what red count is expected?

Hypergeometric standard deviation

Let $N = r + b$ and $p = r/N$. Synthetic urn: $r = 7$ red, $b = 5$ blue; draw $n = 4$ without replacement.

$$\sigma = \sqrt{np(1-p)\frac{N-n}{N-1}}$$

What is the standard deviation of the red count?

Compare sampling designs

A synthetic box has 7 red and 5 blue tokens. Plan A replaces each token; plan B keeps each token.

Which plan permits an exact binomial model?

A small sampling fraction

A random sample without replacement uses a tiny fraction of a large population.

Why might a binomial model be a useful approximation rather than an exact model?

Select the distribution

Select 6 cables without replacement from a known batch and count the faulty ones.

Which discrete distribution matches this sampling design?